

JUSTIN JOE MATHEW

SOFTWARE ENGINEER · SYSTEMS, VECTORIZED SIMULATION & MOBILE ARCHITECTURE

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TECHNICAL SKILLS

Languages	Python, C, C++, Kotlin, TypeScript, JavaScript, Dart, SQL
Systems & Simulation	NumPy Vectorization, Monte Carlo Simulation, Bellman Dynamic Programming, FastF1 Telemetry Engine
Mobile & Wear OS	Android Native, Jetpack Compose, Kotlin Coroutines/Flow, Room SQLite, WorkManager, Wear OS, Flutter
Full-Stack & AI	React, Next.js, Node.js, FastAPI, WebGL, Gemini API, Multi-Agent Orchestration, WebSockets, REST APIs
Tools & Infrastructure	Git, GitHub, Linux Shell, Docker, Vercel, Sensor Telemetry (GPS / Pulse Fusion), CI/CD

FEATURED ENGINEERING PROJECTS

Formula 1 Monte Carlo Race Strategy Simulator Python · NumPy · FastF1 · Dynamic Programming · React
[Live Application ↗](#) | [github.com/jvstin47/f1-monte-carlo-race-strategy-simulator ↗](https://github.com/jvstin47/f1-monte-carlo-race-strategy-simulator)

- Architected a vectorized simulation engine in Python executing **10,000 parallel race iterations in milliseconds** using NumPy arrays to model tyre degradation curves, safety car stochasticity, and fuel-mass lap deltas.
- Formulated a **dynamic programming solver via the Bellman equation** to calculate optimal pit windows and compound selection under rival undercut dynamics and changing track states.
- Engineered an interactive web frontend delivering live telemetry curves, strategy matrices, and pit loss analysis.

AroggyaGram-3 — Intelligent Emergency Health Response Dispatch TypeScript · Next.js · Multi-Agent AI · WebSockets
[github.com/jvstin47/AroggyaGram-3 ↗](https://github.com/jvstin47/AroggyaGram-3)

- Engineered a decentralized crisis dispatch platform converting unstructured voice/text distress calls into rapid field action.
- Integrated Gemini LLM multi-agent pipelines for automated intent classification and medical triage severity categorization, mapping tasks to verified field workers via geo-fenced proximity routing.
- Built zero-dropout architecture with SMS failover routing, guaranteeing critical distress persistence under degraded connectivity.

AutoMeter — Smart Dynamic Transit Meter & Instant UPI Platform Flutter · Dart · Android Native · Sensor Telemetry
[github.com/jvstin47/AutoMeter ↗](https://github.com/jvstin47/AutoMeter) | **Winner — Idea From Industry Season 2**

- Engineered a tamper-resilient transit fare meter fusing GPS speed deltas and wheel pulse tachometer inputs for precise fare calculation under GPS drift and high vibration.
- Automated on-screen dynamic UPI QR generation reflecting real-time fare calculations, eliminating transit payment friction.

Opportunity — Offline-First Android & Wear OS Task OS Kotlin · Jetpack Compose · Room DB · Wear OS
[github.com/jvstin47/opportunity-android-task-planner ↗](https://github.com/jvstin47/opportunity-android-task-planner)

- Built an offline-first life management system featuring deterministic task scoring and capacity-aware schedule allocation.
- Architected with Jetpack Compose (UDF), Room SQLite persistence, WorkManager background jobs, and a Wear OS companion.

HONORS, PODIUMS & HACKATHONS

- 1ST PLACE** **Winner — ACM Vibefest Hackathon 2026** (ACM-W Chapter, March 2026) — Awarded 1st Place in 24-hr engineering hackathon for rapid system architecture, technical performance, and high-impact delivery.
- WINNER** **Winner (Stage 1 & 2) — Idea From Industry Challenge Season 2** (MakerHub IEDC · BIRAC · MoE, Jan–Feb 2026) — 1st place award for smart transit telemetry architecture; advanced through state-level competitive industry evaluations.
- 1ST PLACE** **Winner — AI Club Ideathon 2026** (AJCE Autonomous, Feb 2026) — Secured victory for algorithmic multi-agent crisis response.
- NATIONAL** **Shortlisted — Smart India Hackathon (SIH 2026)** (Ministry of Education Innovation Cell & AICTE) — Certificate of Appreciation for outstanding achievement in nationwide hackathon shortlist.
- GOVT AWARD** **State Innovation Honour — K-DISC Young Innovators Programme (YIP 8.0)** (Govt. of Kerala, Oct 2025) — State recognition for high-potential technological research and innovation.
- GLOBAL** **NASA Space Apps Challenge 2025** (NASA, Oct 2025) — Applied spatial computation and aerospace data pipelines to open NASA datasets.
- PODIUM** **Double 3rd Prize (Podium) — Azure 2026** (Feb 2026) — 3rd Prize in Extempore English & 3rd Prize in JamSpace (Just A Minute).

EDUCATION

Amal Jyothi College of Engineering (Autonomous) Bachelor of Technology (B.Tech) in Computer Science & Engineering	Kerala, India · 2025 – 2029
Fatimagiri Higher Secondary School — CBSE Class XII	Nilambur, Kerala · 2025 · 85.60%
St. Mary's School — CBSE Class X	Safdarjung, New Delhi · 2023 · 81.30%